Gandaki University

Faculty of Science and Technology

Bachelor of Information Technology (BIT)

Project Progress Interaction Log Sheet

Academic Year: 2022 Semester: 8th Visit No: 2 Date of Visit:

Project Title: Dhoom Car Racing

Student Name(s): Aakash Rana Magar, Ayush Tamrakar, Roshan Thapa

Student ID(s):

Student–Supervisor Meeting Log (Minimum 5 interactions are required):

Completed Tasks/ Achievement	Isssues/ challenges	Next Week plan
1. Implemented full engine simulation (EngineController in EngineManager.cs) with RPM calculation via wheel RPM averaging, gear shifting with up/down bounce, torque curves via AnimationCurve, and rev limiter. 2. Implemented steering with speed-dependent angle curve and counter-steer for drift correction (SteerManager.cs). 3. Developed wheel friction management with slip-based stiffness curves for realistic drift behavior (wheelsManager.cs). 4. Created checkpoint system (RaceCheckpoint.cs) with trigger-based detection requiring sequential checkpoint passing. 5. Implemented lap tracking (PlayerLapTracker.cs) with configurable lap count (1, 2, 3, 5 laps), anti-cheat preventing early finish, and missed checkpoint warnings. 6. Built race countdown (RaceManager.cs) with animated 3-2-1-GO sequence using scale pop-in, hold, and fade-out. 7. Implemented anti-roll bar physics (AntiRollBar.cs) to prevent car flipping during sharp turns.	1. Two parallel car physics implementations exist: PhotonCarController.cs (active for multiplayer) and CarStateMachine system (CarController, EngineManager, SteerManager, WheelsManager) causing maintenance confusion. 2. Checkpoint ordering validation required careful edge-case handling for out-of-sequence detection in PlayerLapTracker.cs. 3. DriftFactor in PhotonCarController uses world-space lateral velocity damping which behaves differently at varying frame rates.	1. Integrate Photon PUN2 for real-time multiplayer: room creation (Room_XXXX), room joining by code, and up to 4 players. 2. Implement network car position/rotation synchronization using IPunObservable with lerp/slerp interpolation. 3. Build lobby system with live player list, ready/unready toggle, and host-only start button. 4. Add Google OAuth2 desktop sign-in for player identity in multiplayer.

Supervisor's Feedback:

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Supervisor Name & Signature: Er. Saroj Giri ________________

Date: ____________________